

SCR2043 OPERATING SYSTEMS

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Section : 02

Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to process monitoring and management in operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment02-studentname-matricno.docx**.

Essential Steps Before Starting Lab Assessment 2:

1. Download necessary source codes:

Use the `wget` command to retrieve the following source code files to your Linux (or WSL or MacOS) environment:

```
wget -O mainprocess.c https://rebrand.ly/mainprocess_c  
wget -O subprocess1.c https://rebrand.ly/subprocess1_c  
wget -O subprocess2.c https://rebrand.ly/subprocess2_c
```

2. Compile the source files:

Use the `gcc` compiler to create executable files from the source code.

```
gcc mainprocess.c -o mainprocess  
gcc subprocess1.c -o subprocess1  
gcc subprocess2.c -o subprocess2
```

3. Execute the dummy processes:

Run all the dummy processes

```
./mainprocess &
```

Press **enter** two times.

Lab Assessment 2 : Linux Process Monitoring and Management

root	44	2	0	03:58	?	00:00:00	[irq/9-acpi]
root	45	2	0	03:58	?	00:00:00	[kworker/R-tpm_d]
root	46	2	0	03:58	?	00:00:00	[kworker/R-ata_s]
root	47	2	0	03:58	?	00:00:00	[kworker/R-md]
root	48	2	0	03:58	?	00:00:00	[kworker/R-md_bi]
root	49	2	0	03:58	?	00:00:00	[kworker/R-edac-]
root	50	2	0	03:58	?	00:00:00	[kworker/R-devfr]
root	51	2	0	03:58	?	00:00:00	[watchdogd]
root	52	2	0	03:58	?	00:00:00	[kworker/1:1H-kblockd]
root	53	2	0	03:58	?	00:00:00	[kswapd0]
root	54	2	0	03:58	?	00:00:00	[ecryptfs-kthread]
root	55	2	0	03:58	?	00:00:00	[kworker/R-kthro]
root	56	2	0	03:58	?	00:00:00	[kworker/R-acpi_]
root	57	2	0	03:58	?	00:00:00	[kworker/u6:2-flush-8:0]
root	58	2	0	03:58	?	00:00:00	[kworker/1:2-cgroup_dest]
root	59	2	0	03:58	?	00:00:00	[scsi_eh_0]
root	60	2	0	03:58	?	00:00:00	[kworker/R-scsi_]
root	61	2	0	03:58	?	00:00:00	[scsi_eh_1]
root	62	2	0	03:58	?	00:00:00	[kworker/R-scsi_]
root	63	2	0	03:58	?	00:00:00	[kworker/u6:3-events_unb]
root	64	2	0	03:58	?	00:00:00	[kworker/u6:4-events_unb]
root	65	2	0	03:58	?	00:00:00	[kworker/R-ml]
root	66	2	0	03:58	?	00:00:00	[kworker/R-ipv6_]
root	73	2	0	03:58	?	00:00:00	[kworker/R-kstrp]
root	75	2	0	03:58	?	00:00:00	[kworker/u7:0]
root	76	2	0	03:58	?	00:00:00	[kworker/u8:0]
root	77	2	0	03:58	?	00:00:00	[kworker/u9:0]
root	82	2	0	03:58	?	00:00:00	[kworker/R-crypt]
root	93	2	0	03:58	?	00:00:00	[kworker/R-charge]
root	132	2	0	03:58	?	00:00:01	[kworker/0:2-events]
root	136	2	0	03:58	?	00:00:00	[kworker/u4:1-ext4-rsv-c]
root	137	2	0	03:58	?	00:00:00	[kworker/0:3-events]
root	145	2	0	03:58	?	00:00:00	[kworker/R-mpt_p]
root	152	2	0	03:58	?	00:00:00	[kworker/R-mpt/0]
root	157	2	0	03:58	?	00:00:00	[scsi_eh_2]
root	158	2	0	03:58	?	00:00:00	[kworker/R-scsi_]
root	192	2	0	03:58	?	00:00:00	[kworker/R-raid5]
root	229	2	0	03:58	?	00:00:00	[jbd2/sda1-8]
root	230	2	0	03:58	?	00:00:00	[kworker/R-ext4-]
root	293	2	0	03:58	?	00:00:00	[kworker/0:2H-kblockd]
root	293	2	0	03:58	?	00:00:00	[kworker/0:2H-kblockd]
root	298	2	0	03:58	?	00:00:00	[psimon]
root	303	1	0	03:58	?	00:00:00	/usr/lib/systemd/systemd
root	323	2	0	03:58	?	00:00:00	[kworker/u5:3-events_unb]
root	324	2	0	03:58	?	00:00:00	[kworker/R-kmpat]
root	325	2	0	03:58	?	00:00:00	[kworker/R-kmpat]
root	359	1	0	03:58	?	00:00:00	/sbin/multipathd -d -s
root	366	1	0	03:58	?	00:00:00	/usr/lib/systemd/systemd
root	381	2	0	03:58	?	00:00:00	[psimon]
root	457	2	0	03:58	?	00:00:00	[jbd2/sda16-8]
root	458	2	0	03:58	?	00:00:00	[kworker/R-ext4-]
systemd+	492	1	0	03:58	?	00:00:00	/usr/lib/systemd/systemd
root	558	2	0	03:58	?	00:00:00	[irq/18-vmwgfx]
root	559	2	0	03:58	?	00:00:00	[kworker/R-ttm]
systemd+	562	1	0	03:58	?	00:00:00	/usr/lib/systemd/systemd
systemd+	571	1	0	03:58	?	00:00:00	/usr/lib/systemd/systemd
message+	639	1	0	03:58	?	00:00:00	@dbus-daemon --system --
polkitd	644	1	0	03:58	?	00:00:00	/usr/lib/polkit-1/polkit
root	660	1	0	03:58	?	00:00:00	/usr/lib/systemd/systemd
root	661	1	0	03:58	?	00:00:00	/usr/libexec/udisks2/udi
root	687	2	0	03:58	?	00:00:00	[kworker/u4:2]
root	720	1	0	03:58	?	00:00:00	/usr/bin/python3 /usr/sh
syslog	751	1	0	03:58	?	00:00:00	/usr/sbin/rsyslogd -n -i
root	759	1	0	03:58	?	00:00:00	/usr/sbin/ModemManager
root	827	1	0	03:58	?	00:00:00	/usr/sbin/cron -f -P
root	835	1	0	03:58	tty1	00:00:00	/sbin/agetty -o -p -- \u
root	899	1	0	04:00	?	00:00:00	sshd: /usr/sbin/sshd -D
root	900	899	0	04:00	?	00:00:00	sshd: naim [priv]
naim	902	900	0	04:00	?	00:00:00	sshd: naim@pts/0
naim	903	902	0	04:00	pts/0	00:00:00	-bash
naim	930	903	0	04:01	pts/0	00:00:00	./mainprocess
naim	931	930	0	04:01	pts/0	00:00:00	./mainprocess
naim	932	930	0	04:01	pts/0	00:00:00	./mainprocess
naim	933	931	0	04:01	pts/0	00:00:00	./subprocess1
naim	934	931	0	04:01	pts/0	00:00:00	./subprocess1
naim	935	932	0	04:01	pts/0	00:00:00	./subprocess2
naim	936	932	0	04:01	pts/0	00:00:00	./subprocess2
naim	937	932	0	04:01	pts/0	00:00:00	./subprocess2
naim	938	903	0	04:01	pts/0	00:00:00	./mainprocess
naim	939	938	0	04:01	pts/0	00:00:00	./mainprocess
naim	940	938	0	04:01	pts/0	00:00:00	./mainprocess
naim	941	939	0	04:01	pts/0	00:00:00	./subprocess1
naim	942	939	0	04:01	pts/0	00:00:00	./subprocess1
naim	943	940	0	04:01	pts/0	00:00:00	./subprocess2
naim	944	940	0	04:01	pts/0	00:00:00	./subprocess2
naim	945	940	0	04:01	pts/0	00:00:00	./subprocess2
root	949	2	0	04:04	?	00:00:00	[kworker/u5:0-flush-8:0]
naim	953	903	99	04:05	pts/0	00:00:00	ps -ef

Employ the `ps` command with necessary options to unveil comprehensive details about each running process.

```
Command
ps -eo pid,ppid,user,stat,start,time,cmd

Output
naim@secre2043:~$ ps -eo pid,ppid,user,stat,start,time,cmd
PID  PPID  USER  STAT  STARTED  TIME  CMD
1    0    root   Ss     03:58:20 00:00:00 /sbin/init
2    0    root   S      03:58:20 00:00:00 [kthreadd]
3    2    root   S      03:58:20 00:00:00 [pool_workqueue_release]
4    2    root   I<     03:58:20 00:00:00 [kworker/R-rcu_g]
5    2    root   I<     03:58:20 00:00:00 [kworker/R-rcu_p]
6    2    root   I<     03:58:20 00:00:00 [kworker/R-slab_]
7    2    root   I<     03:58:20 00:00:00 [kworker/R-netns]
10   2    root   I<     03:58:20 00:00:00 [kworker/0:0H-kblockd]
12   2    root   I<     03:58:20 00:00:00 [kworker/R-mm_pe]
13   2    root   I      03:58:20 00:00:00 [rcu_tasks_kthread]
14   2    root   I      03:58:20 00:00:00 [rcu_tasks_rude_kthread]
15   2    root   I      03:58:20 00:00:00 [rcu_tasks_trace_kthread]
16   2    root   S      03:58:20 00:00:00 [ksoftirqd/0]
17   2    root   I      03:58:20 00:00:00 [rcu_preempt]
18   2    root   S      03:58:20 00:00:00 [migration/0]
19   2    root   S      03:58:20 00:00:00 [idle_inject/0]
20   2    root   S      03:58:20 00:00:00 [cpuhp/0]
21   2    root   S      03:58:20 00:00:00 [cpuhp/1]
22   2    root   S      03:58:20 00:00:00 [idle_inject/1]
23   2    root   S      03:58:20 00:00:00 [migration/1]
24   2    root   S      03:58:20 00:00:00 [ksoftirqd/1]
26   2    root   I<     03:58:20 00:00:00 [kworker/1:0H-events_highpri]
28   2    root   I      03:58:20 00:00:00 [kworker/u6:0-events_unbound]
29   2    root   S      03:58:20 00:00:00 [kdevtmpfs]
30   2    root   I<     03:58:20 00:00:00 [kworker/R-inet_]
31   2    root   I      03:58:20 00:00:00 [kworker/u5:1-events_unbound]
32   2    root   S      03:58:20 00:00:00 [kauditd]
33   2    root   S      03:58:20 00:00:00 [khungtaskd]
34   2    root   S      03:58:20 00:00:00 [oom_reaper]
35   2    root   I      03:58:20 00:00:00 [kworker/u6:1-events_unbound]
36   2    root   I<     03:58:20 00:00:00 [kworker/R-write]
37   2    root   S      03:58:20 00:00:00 [kcompactd0]
38   2    root   SN     03:58:20 00:00:00 [ksmd]
39   2    root   I      03:58:20 00:00:05 [kworker/1:1-events]
40   2    root   SN     03:58:20 00:00:00 [khugepaged]
41   2    root   I<     03:58:20 00:00:00 [kworker/R-kint]
42   2    root   I<     03:58:20 00:00:00 [kworker/R-kbloc]
43   2    root   I<     03:58:20 00:00:00 [kworker/R-blkc]
44   2    root   S      03:58:20 00:00:00 [irq/9-acpi]
```

```

44      2 root      S      03:58:20 00:00:00 [irq/9-acpi]
45      2 root      I<     03:58:20 00:00:00 [kworker/R-tpm_d]
46      2 root      I<     03:58:20 00:00:00 [kworker/R-ata_s]
47      2 root      I<     03:58:20 00:00:00 [kworker/R-md]
48      2 root      I<     03:58:20 00:00:00 [kworker/R-md_bi]
49      2 root      I<     03:58:20 00:00:00 [kworker/R-edac-]
50      2 root      I<     03:58:20 00:00:00 [kworker/R-devfr]
51      2 root      S      03:58:20 00:00:00 [watchdogd]
52      2 root      I<     03:58:20 00:00:00 [kworker/1:1H-kblockd]
53      2 root      S      03:58:20 00:00:00 [kswapd0]
54      2 root      S      03:58:20 00:00:00 [ecryptfs-kthread]
55      2 root      I<     03:58:20 00:00:00 [kworker/R-kthro]
56      2 root      I<     03:58:20 00:00:00 [kworker/R-acpi_]
57      2 root      I      03:58:20 00:00:00 [kworker/u6:2-flush-8:0]
58      2 root      I      03:58:20 00:00:00 [kworker/1:2-cgroup_destroy]
59      2 root      S      03:58:20 00:00:00 [scsi_eh_0]
60      2 root      I<     03:58:20 00:00:00 [kworker/R-scsi_]
61      2 root      S      03:58:20 00:00:00 [scsi_eh_1]
62      2 root      I<     03:58:20 00:00:00 [kworker/R-scsi_]
63      2 root      I      03:58:20 00:00:00 [kworker/u6:3-events_unbound]
64      2 root      I      03:58:20 00:00:00 [kworker/u6:4-events_unbound]
65      2 root      I<     03:58:20 00:00:00 [kworker/R-mld]
66      2 root      I<     03:58:20 00:00:00 [kworker/R-ipv6_]
73      2 root      I<     03:58:20 00:00:00 [kworker/R-kstrp]
75      2 root      I<     03:58:20 00:00:00 [kworker/u7:0]
76      2 root      I<     03:58:20 00:00:00 [kworker/u8:0]
77      2 root      I<     03:58:20 00:00:00 [kworker/u9:0]
82      2 root      I<     03:58:20 00:00:00 [kworker/R-crypt]
93      2 root      I<     03:58:20 00:00:00 [kworker/R-charg]
132     2 root      I      03:58:20 00:00:01 [kworker/0:2-ata_sff]
136     2 root      I      03:58:20 00:00:00 [kworker/u4:1-ext4-rsv-conve]
137     2 root      I      03:58:20 00:00:00 [kworker/0:3-events]
145     2 root      I<     03:58:21 00:00:00 [kworker/R-mpt_p]
152     2 root      I<     03:58:21 00:00:00 [kworker/R-mpt/0]
157     2 root      S      03:58:21 00:00:00 [scsi_eh_2]
158     2 root      I<     03:58:21 00:00:00 [kworker/R-scsi_]
192     2 root      I<     03:58:23 00:00:00 [kworker/R-raid5]
229     2 root      S      03:58:24 00:00:00 [jbd2/sda1-8]
230     2 root      I<     03:58:24 00:00:00 [kworker/R-ext4-]
293     2 root      I<     03:58:25 00:00:00 [kworker/0:2H-kblockd]

```

```

293     2 root      I<     03:58:25 00:00:00 [kworker/0:2H-kblockd]
298     2 root      S      03:58:25 00:00:00 [psimon]
303     1 root      S<s    03:58:25 00:00:00 /usr/lib/systemd/systemd-jou
323     2 root      I      03:58:26 00:00:00 [kworker/u5:3-events_unbound]
324     2 root      I<     03:58:26 00:00:00 [kworker/R-kmpat]
325     2 root      I<     03:58:26 00:00:00 [kworker/R-kmpat]
359     1 root      SLSl   03:58:26 00:00:00 /sbin/multipathd -d -s
366     1 root      Ss     03:58:26 00:00:00 /usr/lib/systemd/systemd-ude
381     2 root      S      03:58:26 00:00:00 [psimon]
457     2 root      S      03:58:26 00:00:00 [jbd2/sda16-8]
458     2 root      I<     03:58:26 00:00:00 [kworker/R-ext4-]
492     1 systemd+ Ss     03:58:26 00:00:00 /usr/lib/systemd/systemd-net
558     2 root      S      03:58:27 00:00:00 [irq/18-vmwgfx]
559     2 root      I<     03:58:27 00:00:00 [kworker/R-ttm]
562     1 systemd+ Ss     03:58:27 00:00:00 /usr/lib/systemd/systemd-res
571     1 systemd+ Ssl   03:58:27 00:00:00 /usr/lib/systemd/systemd-tim
639     1 message+ Ss     03:58:27 00:00:00 @dbus-daemon --system --addr
644     1 polkitd     Ssl   03:58:27 00:00:00 /usr/lib/polkit-1/polkitd --
660     1 root      Ss     03:58:27 00:00:00 /usr/lib/systemd/systemd-log
661     1 root      Ssl   03:58:27 00:00:00 /usr/libexec/udisks2/udisksd
687     2 root      I      03:58:27 00:00:00 [kworker/u4:2]
720     1 root      Ssl   03:58:27 00:00:00 /usr/bin/python3 /usr/share/
751     1 syslog     Ssl   03:58:27 00:00:00 /usr/sbin/rsyslogd -n -iNONE
759     1 root      Ssl   03:58:27 00:00:00 /usr/sbin/ModemManager
827     1 root      Ss     03:58:29 00:00:00 /usr/sbin/cron -f -P
835     1 root      Ss+    03:58:29 00:00:00 /sbin/agetty -o -p -- \u --n
899     1 root      Ss     04:00:08 00:00:00 sshd: /usr/sbin/sshd -D [lis
900     899 root      Ss     04:00:08 00:00:00 sshd: naim [priv]
902     900 naim     S      04:00:16 00:00:00 sshd: naim@pts/0
903     902 naim     Ss     04:00:16 00:00:00 -bash
930     903 naim     S      04:01:24 00:00:00 ./mainprocess
931     930 naim     S      04:01:24 00:00:00 ./mainprocess
932     930 naim     S      04:01:24 00:00:00 ./mainprocess
933     931 naim     S      04:01:24 00:00:00 ./subprocess1
934     931 naim     S      04:01:24 00:00:00 ./subprocess1
935     932 naim     S      04:01:24 00:00:00 ./subprocess2
936     932 naim     S      04:01:24 00:00:00 ./subprocess2
937     932 naim     S      04:01:24 00:00:00 ./subprocess2
938     903 naim     S      04:01:54 00:00:00 ./mainprocess
939     938 naim     S      04:01:54 00:00:00 ./mainprocess

```

```

940  938 naim  S   04:01:54 00:00:00 ./mainprocess
941  939 naim  S   04:01:54 00:00:00 ./subprocess1
942  939 naim  S   04:01:54 00:00:00 ./subprocess1
943  940 naim  S   04:01:54 00:00:00 ./subprocess2
944  940 naim  S   04:01:54 00:00:00 ./subprocess2
945  940 naim  S   04:01:54 00:00:00 ./subprocess2
949    2 root   I   04:04:01 00:00:00 [kworker/u5:0-events_unbound
956  903 naim  R+  04:09:06 00:00:00 ps -eo pid,ppid,user,stat,st

```

Question 3

Use the `ps` command with some tools to only list processes named "subprocess" and show some info about them.

Command	
<code>ps -ef grep subprocess grep -v grep</code>	
Output	
<pre> naim@secr2043:~\$ ps -ef grep subprocess grep -v grep naim 933 931 0 04:01 pts/0 00:00:00 ./subprocess1 naim 934 931 0 04:01 pts/0 00:00:00 ./subprocess1 naim 935 932 0 04:01 pts/0 00:00:00 ./subprocess2 naim 936 932 0 04:01 pts/0 00:00:00 ./subprocess2 naim 937 932 0 04:01 pts/0 00:00:00 ./subprocess2 naim 941 939 0 04:01 pts/0 00:00:00 ./subprocess1 naim 942 939 0 04:01 pts/0 00:00:00 ./subprocess1 naim 943 940 0 04:01 pts/0 00:00:00 ./subprocess2 naim 944 940 0 04:01 pts/0 00:00:00 ./subprocess2 naim 945 940 0 04:01 pts/0 00:00:00 ./subprocess2 </pre>	

Question 4

Execute the `ps` command, specifying options that reveal only the following columns:

- Process ID (`pid`)
- Owner of the process (`user`)
- CPU percentage (`pcpu`)
- Memory percentage (`pmem`)
- Command (`cmd`)

Command
<code>ps -eo pid,user,pcpu,pmem,cmd</code>
Output

```
naim@secre2043:~$ ps -eo pid,user,pcpu,pmem,cmd
PID USER      %CPU %MEM    CMD
  1 root         0.0   1.3  /sbin/init
  2 root         0.0   0.0  [kthreadd]
  3 root         0.0   0.0  [pool_workqueue_release]
  4 root         0.0   0.0  [kworker/R-rcu_g]
  5 root         0.0   0.0  [kworker/R-rcu_p]
  6 root         0.0   0.0  [kworker/R-slub_]
  7 root         0.0   0.0  [kworker/R-netns]
 10 root         0.0   0.0  [kworker/0:0H-kblockd]
 12 root         0.0   0.0  [kworker/R-mm_pe]
 13 root         0.0   0.0  [rcu_tasks_kthread]
 14 root         0.0   0.0  [rcu_tasks_rude_kthread]
 15 root         0.0   0.0  [rcu_tasks_trace_kthread]
 16 root         0.0   0.0  [ksoftirqd/0]
 17 root         0.0   0.0  [rcu_preempt]
 18 root         0.0   0.0  [migration/0]
 19 root         0.0   0.0  [idle_inject/0]
 20 root         0.0   0.0  [cpuhp/0]
 21 root         0.0   0.0  [cpuhp/1]
 22 root         0.0   0.0  [idle_inject/1]
 23 root         0.0   0.0  [migration/1]
 24 root         0.0   0.0  [ksoftirqd/1]
 26 root         0.0   0.0  [kworker/1:0H-events_highpri]
 28 root         0.0   0.0  [kworker/u6:0-events_power_efficient]
 29 root         0.0   0.0  [kdevtmpfs]
 30 root         0.0   0.0  [kworker/R-inet_]
 31 root         0.0   0.0  [kworker/u5:1-events_unbound]
 32 root         0.0   0.0  [kauditd]
 33 root         0.0   0.0  [khungtaskd]
 34 root         0.0   0.0  [oom_reaper]
 35 root         0.0   0.0  [kworker/u6:1-flush-8:0]
 36 root         0.0   0.0  [kworker/R-write]
 37 root         0.0   0.0  [kcompactd0]
 38 root         0.0   0.0  [ksmd]
 39 root         0.8   0.0  [kworker/1:1-events]
 40 root         0.0   0.0  [khugepaged]
 41 root         0.0   0.0  [kworker/R-kinte]
 42 root         0.0   0.0  [kworker/R-kbloc]
 43 root         0.0   0.0  [kworker/R-blkcq]
 44 root         0.0   0.0  [irq/9-acpi]
```

```

44 root      0.0  0.0 [irq/9-acpi]
45 root      0.0  0.0 [kworker/R-tpm_d]
46 root      0.0  0.0 [kworker/R-ata_s]
47 root      0.0  0.0 [kworker/R-md]
48 root      0.0  0.0 [kworker/R-md_bi]
49 root      0.0  0.0 [kworker/R-edac-]
50 root      0.0  0.0 [kworker/R-devfr]
51 root      0.0  0.0 [watchdogd]
52 root      0.0  0.0 [kworker/1:1H-kblockd]
53 root      0.0  0.0 [kswapd0]
54 root      0.0  0.0 [ecryptfs-kthread]
55 root      0.0  0.0 [kworker/R-kthro]
56 root      0.0  0.0 [kworker/R-acpi_]
58 root      0.0  0.0 [kworker/1:2-cgroup_destroy]
59 root      0.0  0.0 [scsi_eh_0]
60 root      0.0  0.0 [kworker/R-scsi_]
61 root      0.0  0.0 [scsi_eh_1]
62 root      0.0  0.0 [kworker/R-scsi_]
63 root      0.0  0.0 [kworker/u6:3-events_power_efficient]
65 root      0.0  0.0 [kworker/R-mld]
66 root      0.0  0.0 [kworker/R-ipv6_]
73 root      0.0  0.0 [kworker/R-kstrp]
75 root      0.0  0.0 [kworker/u7:0]
76 root      0.0  0.0 [kworker/u8:0]
77 root      0.0  0.0 [kworker/u9:0]
82 root      0.0  0.0 [kworker/R-crypt]
93 root      0.0  0.0 [kworker/R-charg]
132 root     0.1  0.0 [kworker/0:2-events]
136 root     0.0  0.0 [kworker/u4:1-ext4-rsv-conversion]
137 root     0.0  0.0 [kworker/0:3-cgroup_destroy]
145 root     0.0  0.0 [kworker/R-mpt_p]
152 root     0.0  0.0 [kworker/R-mpt/0]
157 root     0.0  0.0 [scsi_eh_2]
158 root     0.0  0.0 [kworker/R-scsi_]
192 root     0.0  0.0 [kworker/R-raid5]
229 root     0.0  0.0 [jbd2/sda1-8]
230 root     0.0  0.0 [kworker/R-ext4-]
293 root     0.0  0.0 [kworker/0:2H-kblockd]
298 root     0.0  0.0 [psimon]
303 root     0.0  1.2 /usr/lib/systemd/systemd-journald
323 root     0.0  0.0 [kworker/u5:3-events_unbound]

```



```

323 root      0.0  0.0 [kworker/u5:3-events_unbound]
324 root      0.0  0.0 [kworker/R-kmpat]
325 root      0.0  0.0 [kworker/R-kmpat]
359 root      0.0  2.7 /sbin/multipathd -d -s
366 root      0.0  0.7 /usr/lib/systemd/systemd-udev
381 root      0.0  0.0 [psimon]
457 root      0.0  0.0 [jbd2/sda16-8]
458 root      0.0  0.0 [kworker/R-ext4-]
492 systemd+  0.0  0.9 /usr/lib/systemd/systemd-networkd
558 root      0.0  0.0 [irq/18-vmwgfx]
559 root      0.0  0.0 [kworker/R-ttm]
562 systemd+  0.0  1.3 /usr/lib/systemd/systemd-resolved
571 systemd+  0.0  0.7 /usr/lib/systemd/systemd-timesyncd
639 message+  0.0  0.5 @dbus-daemon --system --address=systemd: --nofork
644 polkitd    0.0  0.7 /usr/lib/polkit-1/polkitd --no-debug
660 root      0.0  0.8 /usr/lib/systemd/systemd-logind
661 root      0.0  1.3 /usr/libexec/udisks2/udisksd
687 root      0.0  0.0 [kworker/u4:2]
720 root      0.0  2.3 /usr/bin/python3 /usr/share/unattended-upgrades/u
751 syslog    0.0  0.5 /usr/sbin/rsyslogd -n -iNONE
759 root      0.0  1.2 /usr/sbin/ModemManager
827 root      0.0  0.2 /usr/sbin/cron -f -P
835 root      0.0  0.1 /sbin/agetty -o -p -- \u --noclear - linux
899 root      0.0  0.8 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 st
900 root      0.0  0.5 sshd: naim [priv]
902 naim       0.0  0.7 sshd: naim@pts/0
903 naim       0.0  0.5 -bash
930 naim       0.0  0.1 ./mainprocess
931 naim       0.0  0.0 ./mainprocess
932 naim       0.0  0.0 ./mainprocess
933 naim       0.0  0.1 ./subprocess1
934 naim       0.0  0.1 ./subprocess1
935 naim       0.0  0.1 ./subprocess2
936 naim       0.0  0.1 ./subprocess2
937 naim       0.0  0.1 ./subprocess2
938 naim       0.0  0.1 ./mainprocess
939 naim       0.0  0.0 ./mainprocess
940 naim       0.0  0.0 ./mainprocess
941 naim       0.0  0.1 ./subprocess1
942 naim       0.0  0.1 ./subprocess1
943 naim       0.0  0.1 ./subprocess2
943 naim       0.0  0.1 ./subprocess2
944 naim       0.0  0.1 ./subprocess2
945 naim       0.0  0.1 ./subprocess2
949 root      0.0  0.0 [kworker/u5:0-events_unbound]
957 root      0.0  0.0 [kworker/u6:2-events_unbound]
960 root      0.0  0.0 [kworker/0:0]
964 naim       0.0  0.4 ps -eo pid,user,pcpu,pmem,cmd

```

Question 5

Building on the ps command used in Question 4, can you add an option to sort the listed processes by their memory usage (pmem)?

Command
ps -eo pid,user,pcpu,pmem,cmd --sort=-pmem
Output

```

naim@secl2043:~$ ps -eo pid,user,pcpu,pmem,cmd --sort=-pmem
PID USER      %CPU %MEM    CMD
359 root        0.0  2.7  /sbin/multipathd -d -s
720 root        0.0  2.3  /usr/bin/python3 /usr/share/unattended-upgrades/u
661 root        0.0  1.3  /usr/libexec/udisks2/udisksd
  1 root        0.0  1.3  /sbin/init
562 systemd+   0.0  1.3  /usr/lib/systemd/systemd-resolved
759 root        0.0  1.2  /usr/sbin/ModemManager
303 root        0.0  1.2  /usr/lib/systemd/systemd-journald
492 systemd+   0.0  0.9  /usr/lib/systemd/systemd-networkd
660 root        0.0  0.8  /usr/lib/systemd/systemd-logind
899 root        0.0  0.8  sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 st
644 polkitd     0.0  0.7  /usr/lib/polkit-1/polkitd --no-debug
571 systemd+   0.0  0.7  /usr/lib/systemd/systemd-timesyncd
366 root        0.0  0.7  /usr/lib/systemd/systemd-udev
902 naim        0.0  0.7  sshd: naim@pts/0
751 syslog     0.0  0.5  /usr/sbin/rsyslogd -n -iNONE
639 message+   0.0  0.5  @dbus-daemon --system --address=systemd: --nofork
900 root        0.0  0.5  sshd: naim [priv]
903 naim        0.0  0.5  -bash
972 naim        0.0  0.4  ps -eo pid,user,pcpu,pmem,cmd --sort=-pmem
827 root        0.0  0.2  /usr/sbin/cron -f -P
835 root        0.0  0.1  /sbin/agetty -o -p -- \u --noclear - linux
930 naim        0.0  0.1  ./mainprocess
933 naim        0.0  0.1  ./subprocess1
934 naim        0.0  0.1  ./subprocess1
935 naim        0.0  0.1  ./subprocess2
936 naim        0.0  0.1  ./subprocess2
937 naim        0.0  0.1  ./subprocess2
938 naim        0.0  0.1  ./mainprocess
941 naim        0.0  0.1  ./subprocess1
942 naim        0.0  0.1  ./subprocess1
943 naim        0.0  0.1  ./subprocess2
944 naim        0.0  0.1  ./subprocess2
945 naim        0.0  0.1  ./subprocess2
931 naim        0.0  0.0  ./mainprocess
932 naim        0.0  0.0  ./mainprocess
939 naim        0.0  0.0  ./mainprocess
940 naim        0.0  0.0  ./mainprocess
  2 root        0.0  0.0  [kthreadd]
  3 root        0.0  0.0  [pool_workqueue_release]

```

```
3 root      0.0  0.0 [pool_workqueue_release]
4 root      0.0  0.0 [kworker/R-rcu_g]
5 root      0.0  0.0 [kworker/R-rcu_p]
6 root      0.0  0.0 [kworker/R-slub_]
7 root      0.0  0.0 [kworker/R-netns]
10 root     0.0  0.0 [kworker/0:0H-kblockd]
12 root     0.0  0.0 [kworker/R-mm_pe]
13 root     0.0  0.0 [rcu_tasks_kthread]
14 root     0.0  0.0 [rcu_tasks_rude_kthread]
15 root     0.0  0.0 [rcu_tasks_trace_kthread]
16 root     0.0  0.0 [ksoftirqd/0]
17 root     0.0  0.0 [rcu_preempt]
18 root     0.0  0.0 [migration/0]
19 root     0.0  0.0 [idle_inject/0]
20 root     0.0  0.0 [cpuhp/0]
21 root     0.0  0.0 [cpuhp/1]
22 root     0.0  0.0 [idle_inject/1]
23 root     0.0  0.0 [migration/1]
24 root     0.0  0.0 [ksoftirqd/1]
26 root     0.0  0.0 [kworker/1:0H-events_highpri]
29 root     0.0  0.0 [kdevtmpfs]
30 root     0.0  0.0 [kworker/R-inet_]
31 root     0.0  0.0 [kworker/u5:1-events_unbound]
32 root     0.0  0.0 [kauditd]
33 root     0.0  0.0 [khungtaskd]
34 root     0.0  0.0 [oom_reaper]
35 root     0.0  0.0 [kworker/u6:1-events_power_efficient]
36 root     0.0  0.0 [kworker/R-write]
37 root     0.0  0.0 [kcompactd0]
38 root     0.0  0.0 [ksmd]
39 root     0.8  0.0 [kworker/1:1-ata_sff]
40 root     0.0  0.0 [khugepaged]
41 root     0.0  0.0 [kworker/R-kinte]
42 root     0.0  0.0 [kworker/R-kbloc]
43 root     0.0  0.0 [kworker/R-blkcg]
44 root     0.0  0.0 [irq/9-acpi]
45 root     0.0  0.0 [kworker/R-tpm_d]
46 root     0.0  0.0 [kworker/R-ata_s]
47 root     0.0  0.0 [kworker/R-md]
48 root     0.0  0.0 [kworker/R-md_bi]
49 root     0.0  0.0 [kworker/R-edac-]
```

```

49 root      0.0  0.0 [kworker/R-edac-]
50 root      0.0  0.0 [kworker/R-devfr]
51 root      0.0  0.0 [watchdogd]
52 root      0.0  0.0 [kworker/1:1H-kblockd]
53 root      0.0  0.0 [kswapd0]
54 root      0.0  0.0 [ecryptfs-kthread]
55 root      0.0  0.0 [kworker/R-kthrd]
56 root      0.0  0.0 [kworker/R-acpi_]
58 root      0.0  0.0 [kworker/1:2-cgroup_destroy]
59 root      0.0  0.0 [scsi_eh_0]
60 root      0.0  0.0 [kworker/R-scsi_]
61 root      0.0  0.0 [scsi_eh_1]
62 root      0.0  0.0 [kworker/R-scsi_]
63 root      0.0  0.0 [kworker/u6:3-events_power_efficient]
65 root      0.0  0.0 [kworker/R-mld]
66 root      0.0  0.0 [kworker/R-ipv6_]
73 root      0.0  0.0 [kworker/R-kstrp]
75 root      0.0  0.0 [kworker/u7:0]
76 root      0.0  0.0 [kworker/u8:0]
77 root      0.0  0.0 [kworker/u9:0]
82 root      0.0  0.0 [kworker/R-crypt]
93 root      0.0  0.0 [kworker/R-charg]
132 root     0.1  0.0 [kworker/0:2-events]
136 root     0.0  0.0 [kworker/u4:1-ext4-rsv-conversion]
137 root     0.0  0.0 [kworker/0:3-cgroup_destroy]
145 root     0.0  0.0 [kworker/R-mpt_p]
152 root     0.0  0.0 [kworker/R-mpt/0]
157 root     0.0  0.0 [scsi_eh_2]
158 root     0.0  0.0 [kworker/R-scsi_]
192 root     0.0  0.0 [kworker/R-raid5]
229 root     0.0  0.0 [jbd2/sda1-8]
230 root     0.0  0.0 [kworker/R-ext4-]
293 root     0.0  0.0 [kworker/0:2H-kblockd]
298 root     0.0  0.0 [psimon]
323 root     0.0  0.0 [kworker/u5:3-events_unbound]
324 root     0.0  0.0 [kworker/R-kmpat]
325 root     0.0  0.0 [kworker/R-kmpat]
381 root     0.0  0.0 [psimon]
457 root     0.0  0.0 [jbd2/sda16-8]
458 root     0.0  0.0 [kworker/R-ext4-]
558 root     0.0  0.0 [irq/18-vmwgfx]

559 root     0.0  0.0 [kworker/R-ttm]
687 root     0.0  0.0 [kworker/u4:2]
949 root     0.0  0.0 [kworker/u5:0-events_unbound]
957 root     0.0  0.0 [kworker/u6:2-events_unbound]
960 root     0.0  0.0 [kworker/0:0]

```

Question 6

Construct a command using `ps`, suitable options, and any additional tools to visualize the hierarchical structure (tree-like) of the following processes:

- "mainprocess"
- "subprocess1"
- "subprocess2"

Command
<code>ps -ejH grep -E "mainprocess subprocess1 subprocess2"</code>

Output					
<pre>naim@secr2043:~\$ ps -ejH grep -E "mainprocess subprocess1 subprocess2"</pre>					
930	930	903	pts/0	00:00:00	mainprocess
931	930	903	pts/0	00:00:00	mainprocess
933	930	903	pts/0	00:00:00	subprocess1
934	930	903	pts/0	00:00:00	subprocess1
932	930	903	pts/0	00:00:00	mainprocess
935	930	903	pts/0	00:00:00	subprocess2
936	930	903	pts/0	00:00:00	subprocess2
937	930	903	pts/0	00:00:00	subprocess2
938	938	903	pts/0	00:00:00	mainprocess
939	938	903	pts/0	00:00:00	mainprocess
941	938	903	pts/0	00:00:00	subprocess1
942	938	903	pts/0	00:00:00	subprocess1
940	938	903	pts/0	00:00:00	mainprocess
943	938	903	pts/0	00:00:00	subprocess2
944	938	903	pts/0	00:00:00	subprocess2
945	938	903	pts/0	00:00:00	subprocess2

Question 7

Use pstree command with option that show the number of threads to each process.

Command	
pstree -p -t	
Output	
<pre>naim@secr2043:~\$ pstree -p -t systemd(1)-+-ModemManager(759)-+-{gdbus}(773) -{gmain}(764) \- {pool-spawner}(768) +-agetty(835) +-cron(827) +-dbus-daemon(639) +-multipathd(359)-+-{multipathd}(373) -{multipathd}(375) -{multipathd}(376) -{multipathd}(377) -{multipathd}(378) \- {multipathd}(379) +-polkitd(644)-+-{gdbus}(748) -{gmain}(746) \- {pool-spawner}(747) +-rsyslogd(751)-+-{in:imklog}(779) -{in:imuxsock}(777) \- {rs:main Q:Reg}(778) +-sshd(899)---sshd(900)---sshd(902)---bash(903)-+-mainprocess(93+ -mainprocess(93+ \-pstree(980) +-systemd-journal(303) +-systemd-logind(660) +-systemd-network(492) +-systemd-resolve(562) +-systemd-timesyn(571)---{sd-resolve}(604) +-systemd-udev(366) +-udisksd(661)-+-{cleanup}(762) -{gdbus}(715) -{gmain}(704) -{pool-spawner}(709) \- {probing-thread}(758) +-unattended-upgr(720)---{gmain}(814)</pre>	

Question 8

Use `renice` command to change priority level of one of process “subprocess1”.

Command
<pre>ps -ef grep subprocess1 sudo renice -n 10 -p 933</pre>
Output
<pre>naim@secre2043:~\$ ps -ef grep subprocess1 naim 933 931 0 04:01 pts/0 00:00:00 ./subprocess1 naim 934 931 0 04:01 pts/0 00:00:00 ./subprocess1 naim 941 939 0 04:01 pts/0 00:00:00 ./subprocess1 naim 942 939 0 04:01 pts/0 00:00:00 ./subprocess1 naim 986 903 0 04:20 pts/0 00:00:00 grep --color=auto subpro cess1 naim@secre2043:~\$ sudo renice -n 10 -p 933 [sudo] password for naim: 933 (process ID) old priority 0, new priority 10</pre>

Question 9

Terminate all running processes with the name “mainprocess”.

Command
<pre>kill -f mainprocess</pre>
Output
<pre>naim@secre2043:~\$ pkill -f mainprocess Main process (ID: 931) received signal: 15. Terminating... Main process (ID: 930) received signal: 15. Terminating... Main process (ID: 938) received signal: 15. Terminating... Main process (ID: 939) received signal: 15. Terminating... Main process (ID: 932) received signal: 15. Terminating... Main process (ID: 940) received signal: 15. Terminating... [1]- Done ./mainprocess [2]+ Done ./mainprocess</pre>

Question 10

Write a short C or Python code (choose only one language) demonstrating multiprocessing with `fork()` and `wait()`. Compile and/or run the code. Show the output.

Source Code:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if (pid < 0) {
        // Fork failed
        perror("fork failed");
        return 1;
    }
    else if (pid == 0) {
        printf("[Child] PID %d is running.\n", getpid());
        sleep(2);
        printf("[Child] PID %d has finished.\n", getpid());
    }

    else {
        printf("[Parent] PID %d created child PID %d.\n", getpid(), pid);
        int status;
        wait(&status);
        printf("[Parent] Child PID %d exited with status %d.\n", pid, status);
    }

    return 0;
}
```

Output:

```
naim@secr2043:~$ nano fork_demo.c
naim@secr2043:~$ gcc fork_demo.c -o fork_demo
naim@secr2043:~$ ./fork_demo
[Parent] PID 1011 created child PID 1012.
[Child] PID 1012 is running.
[Child] PID 1012 has finished.
[Parent] Child PID 1012 exited with status 0.
```